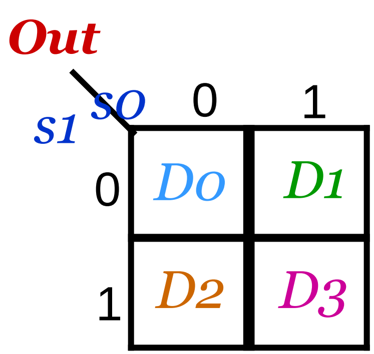
Digital Circuits and Systems

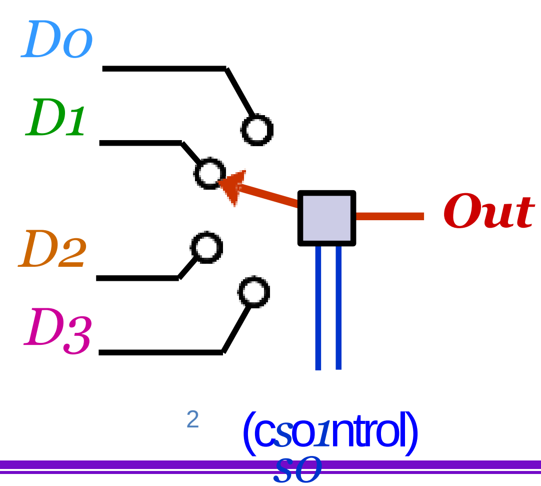
Multiplexers

Multiplexers

- *Multiplexing* means transmitting a large number of information units over a smaller number of channels or lines.
- A *digital multiplexer* (*MUX*) selects binary information from one of many input lines and directs it to a single output line.
  - Data selector(2^n:1 MUX).
  - Inputs: 2^n data inputs, *n* select lines.
  - Output: 1 data output line.

s1	*s0*	*Out*
0	0	*D0*
0	1	*D1*
1	0	*D2*
1	1	*D3*





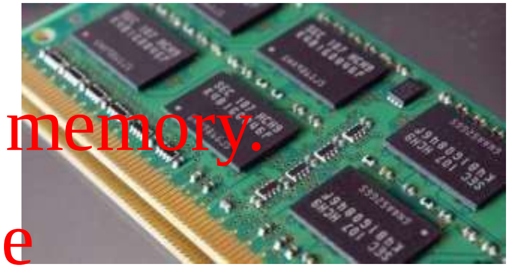
APPLICATIONS OF MULTIPLEXER

❑ **Communication System:**

- ❖ Acommunication multiplexer.

- ❖ By using a multiplexer, the efficiency can be increased.

❑ **Computer Memory:**



- ❖ Multiplexers are used in computer memory.
- ❖ Reduces copper lines to connect the memory to other parts of the computer.

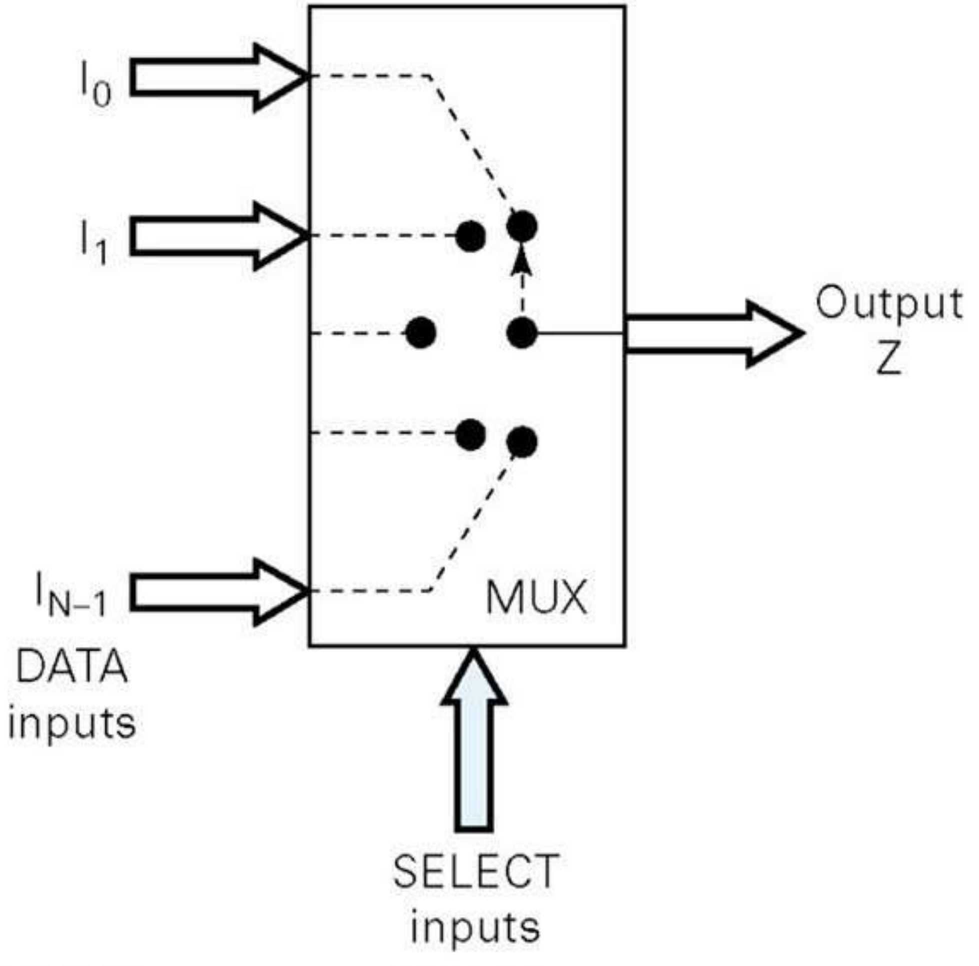
Telephone Network:

❖ In telephone networks, multiple audio signals are integrated.

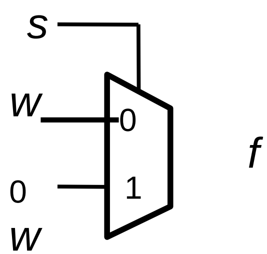
<u>Transmission from the computer of a satellite</u>

Multiplexer is used to transmit the data signal. # from satellite to ground.
by using GSM satellite.

Functional Diagram Of a Multiplexer



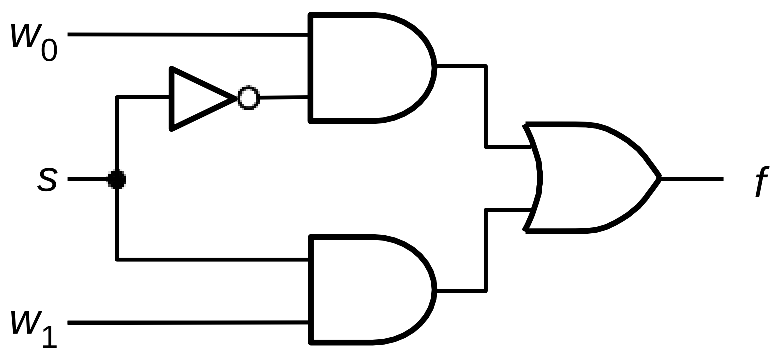
2:1 Mux



(a) Graphical symbol

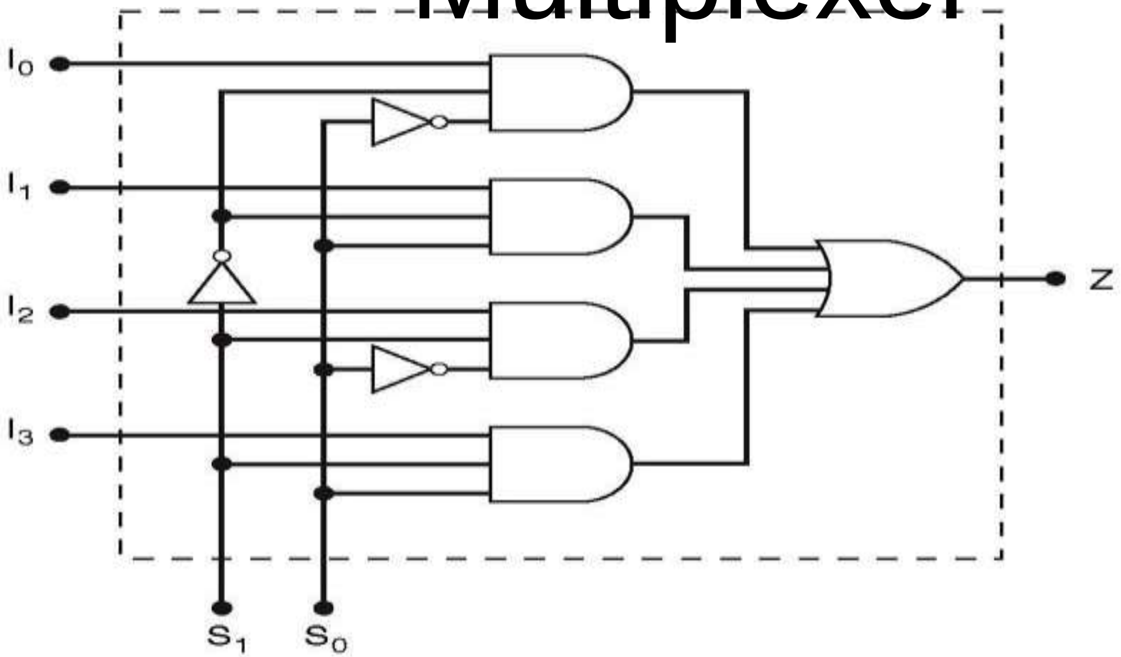
s	f
0	w_0
1	w_1

(b) Truth table



(c) Sum-of-products circuit

4 : 1 Multiplexer

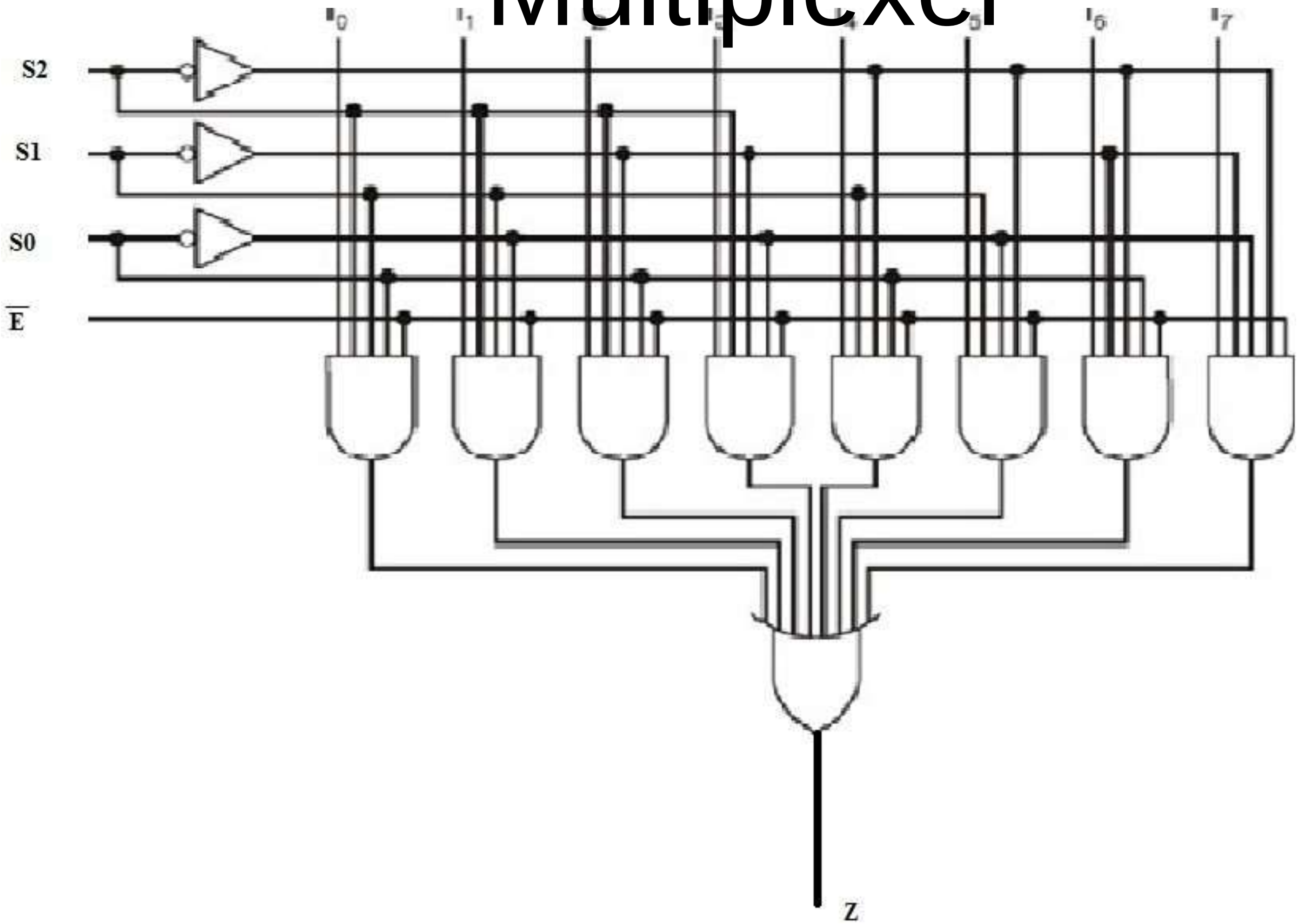


S_0	S_1	Z
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

- A 2^n:1 MUX needs 2^n, (n+1)-input AND gates for selection and a 2^n-input OR gate to generate the final output.
- ⇨ *AND/OR logic structure*

S_0	S_1	Z
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

8 : 1 Multiplexer

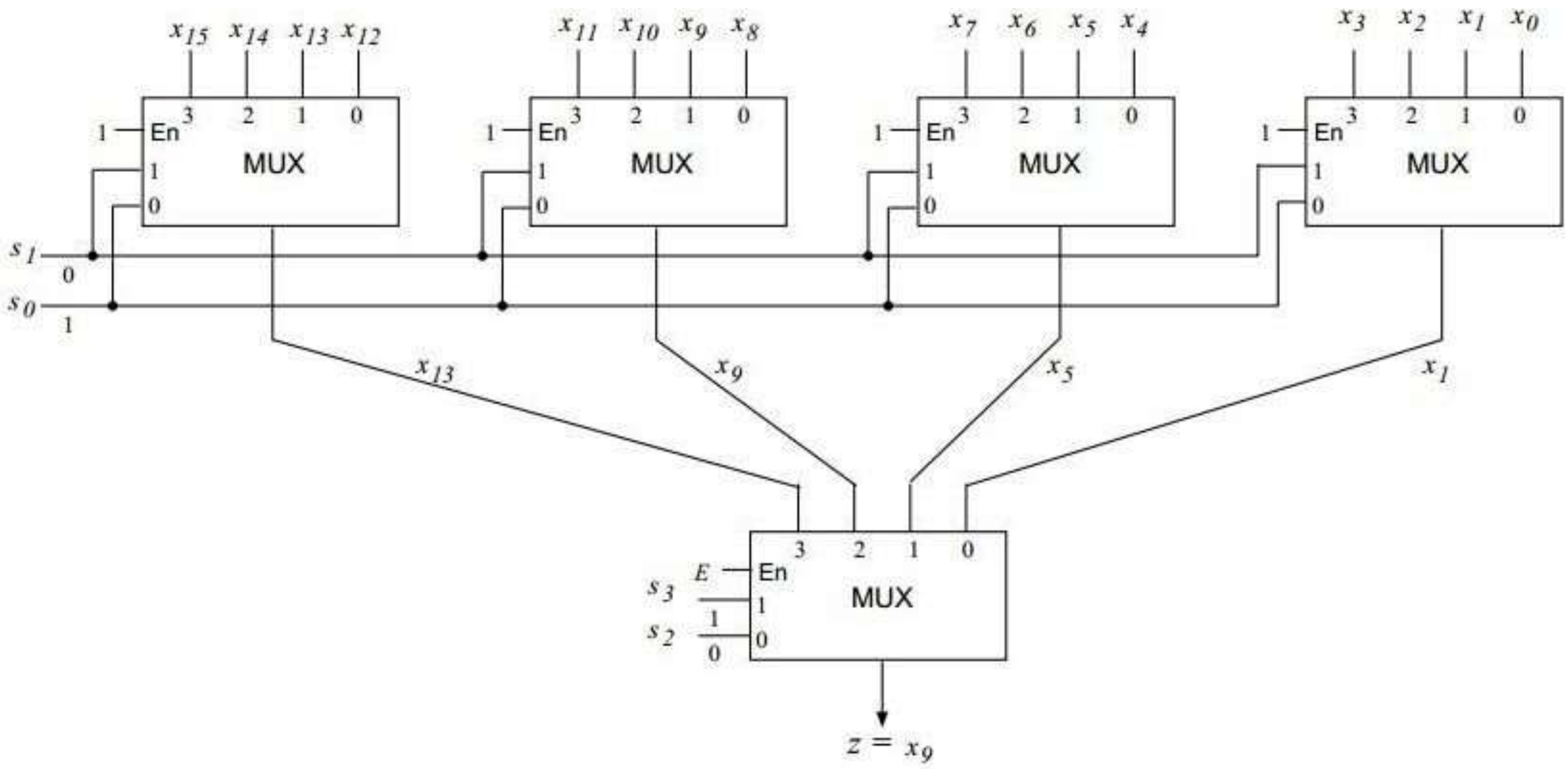


S_0	S_1	S_3	Z
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Multiplexer TREE

- The Multiplexers with more number of inputs can be obtained by cascading two or more multiplexers with less number of
- inputs.

Below is a design of 16:1 MUX using 4 4:1 MUXs :-



Demultiplexer (Data Distributor)

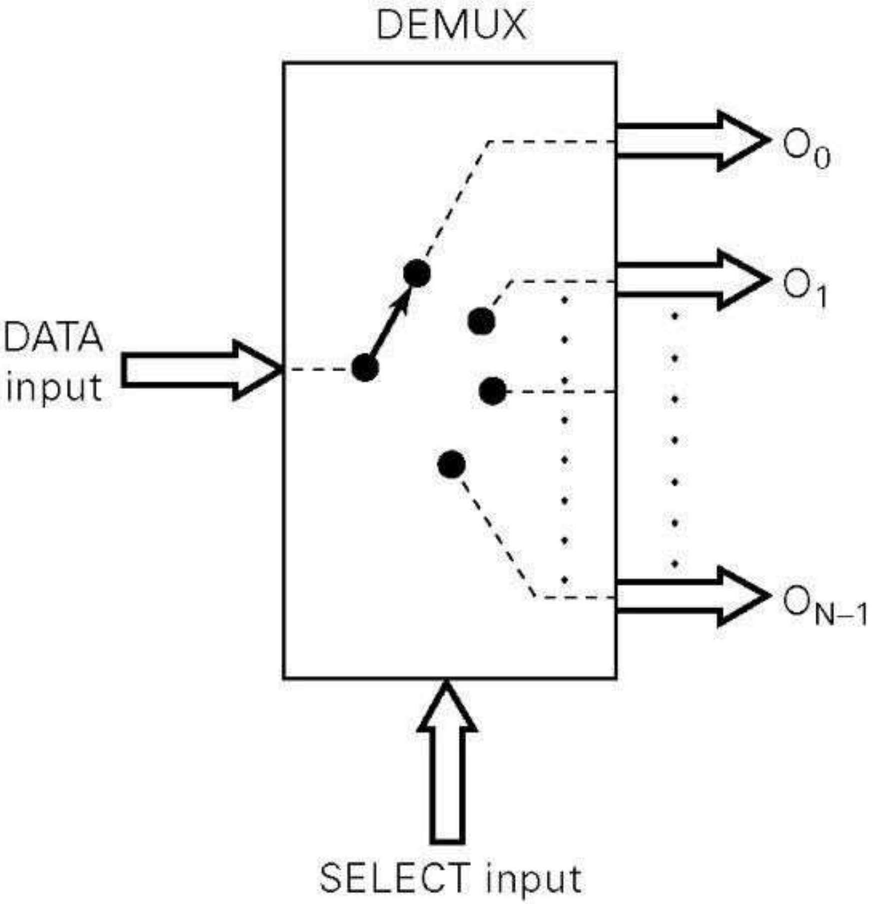
- Definition : A DEMULTIPLEXER (DEMUX) basically reverses the multiplexing function. It takes data from one line and distributes them to a given number of outputs. For this reason, the demultiplexer is also known as a data distributor.

- Single data input lines
- Some select line (less than the no. of output
- lines) Several output line

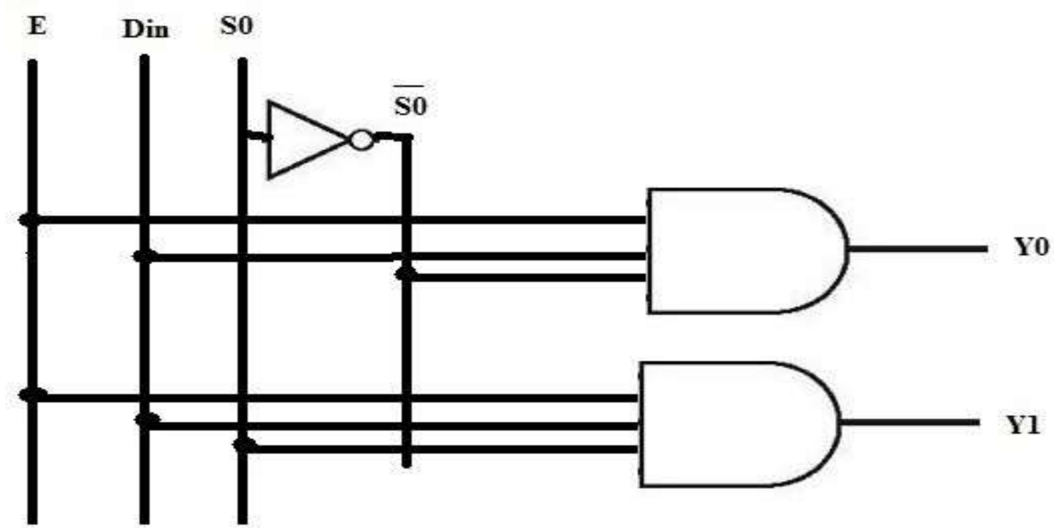
- If there are n data output lines and m select lines, then

$$2^m = n$$

Functional Diagram Of a Demultiplexer

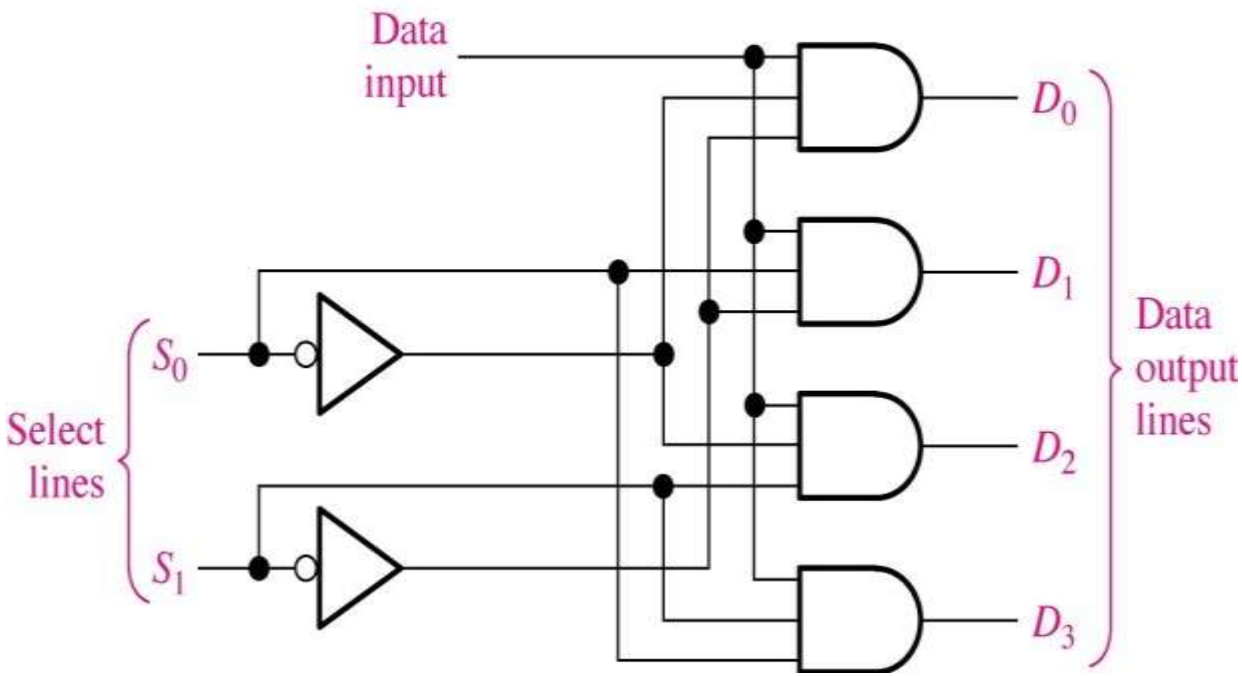


1 : 2 Demultiplexer



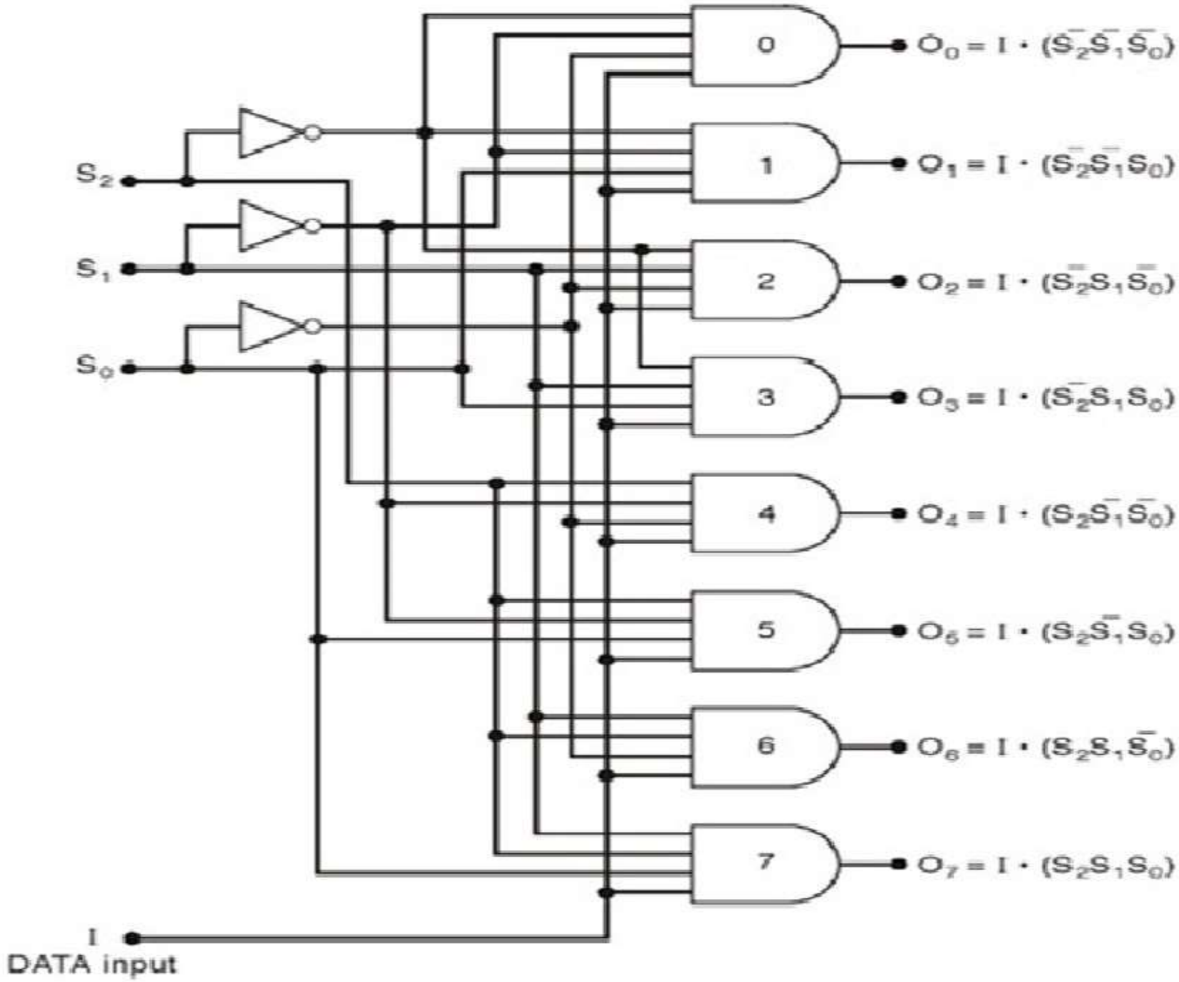
S_0	Y_0	Y_1
1	D	0
2	0	D

1 : 4



S_0	S_1	D_0	D_1	D_2	D_3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D

1 : 8 Demultiplexer



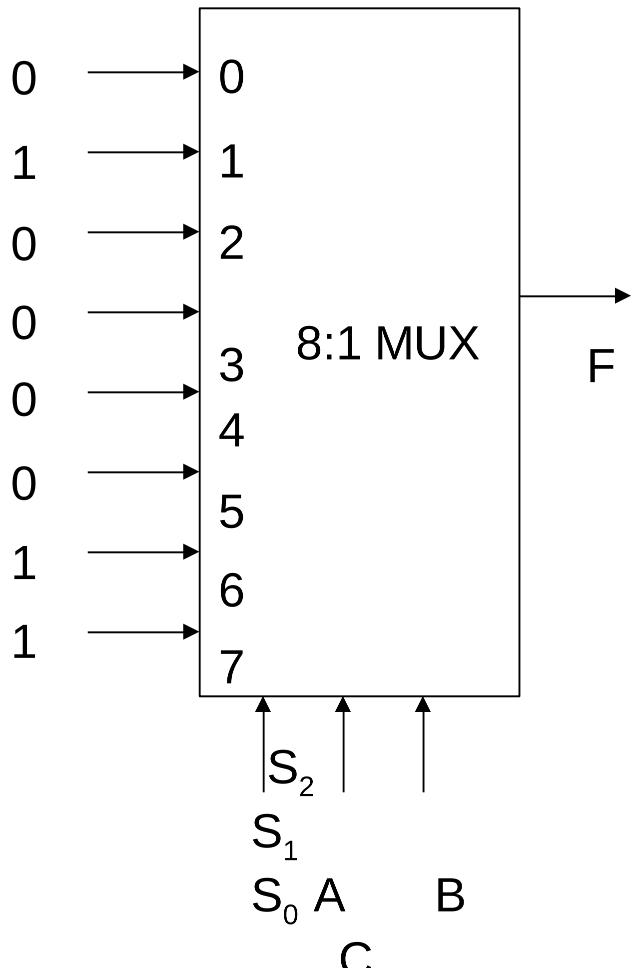
1 : 8 Demultiplexer (Truth Table)

S_0	S_1	S_3	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	D	0	0	0	0	0	0	0
0	0	1	0	D	0	0	0	0	0	0
0	1	0	0	0	D	0	0	0	0	0
0	1	1	0	0	0	D	0	0	0	0
1	0	0	0	0	0	0	D	0	0	0
1	0	1	0	0	0	0	0	D	0	0
1	1	0	0	0	0	0	0	0	D	0
1	1	1	0	0	0	0	0	0	0	D

Implementation Of Logic Functions using Multiplexer

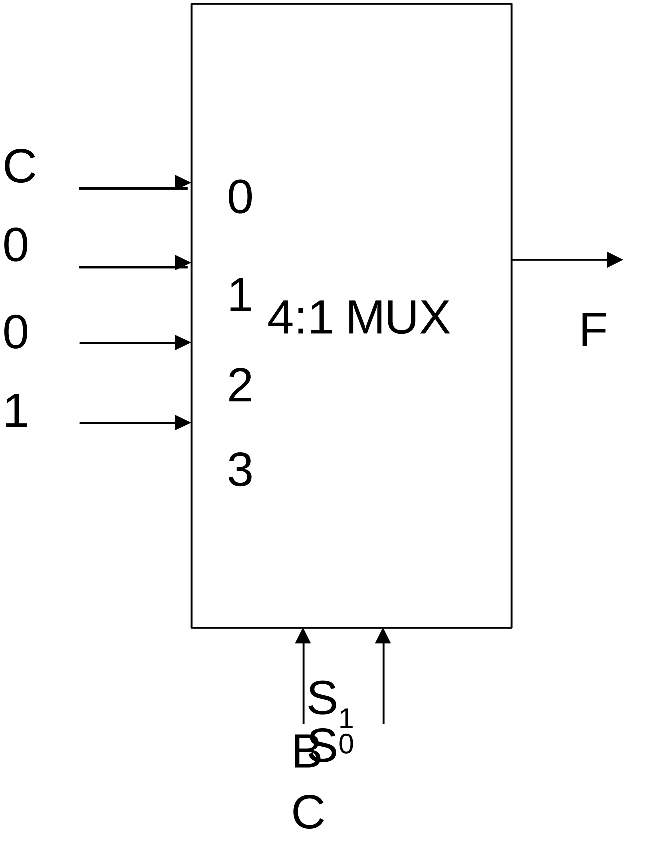
$f(a, b, c) = a'b'c + ab$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



f(a, b, c) = a’b’c + ab

A	B	C	O	F
0	0	0	0	C
0	0	1	1	
0	1	0	0	0
0	1	1	0	
1	0	0	0	0
1	0	1	0	
1	1	0	1	1
1	1	1	1	



THANK
YOU